L9-10: Postulates of Quantum Mechanics, Linear Algebra (Ket, Bra, Operators)

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This session

Postulates of Quantum Mechanics

Kets, Bras

Operators: Hermitian and Unitary Operators

Superposition, Entanglement

Foundation: Postulates of Quantum Mechanics

- A system is represented by a wavefunction
  - Hilbert space
- Evolution is governed by Schrödinger equation
  - Hamiltonian
- Measurement
  - Hermitian operators
- **Superposition**
- **Entanglement**

Interference

For a spin-½ qubit $\mathcal{H} = \mathbb{C}^2$

Basis states: $\{|\uparrow\rangle, |\downarrow\rangle\}$

$$|\psi\rangle = \alpha|\uparrow\rangle + \beta|\downarrow\rangle, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1$$

$$|\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Hermitian Operators (Observables) $\hat{S}_i = \frac{\hbar}{2}\sigma_i$

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

General Hermitian Operator: $\hat{O} = a_0\mathbb{I} + \vec{a} \cdot \vec{\sigma}$

Hamiltonian of a Two-Level Spin Qubit $\hat{H} = \frac{1}{2}\hbar\vec{\Omega} \cdot \vec{\sigma}$

Kets, Bras, Operators

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

- **A Ket is a vector in a complex vector space. Notation:** $|\psi\rangle$
- Invented by Paul Dirac: "Bracket" → Bra + Ket

Example: Classical Bit vs Quantum Bit

- Classical bit: 0 or 1
- Quantum bit (qubit):
- These are basis vectors.

General Qubit State: $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$

Where: $\alpha, \beta \in C$**.** $|\alpha|^2 + |\beta|^2 = 1$**.** This normalization ensures: Total probability = 1.

Kets, Bras, Operators

What is a Ket?

- A Ket is a vector in a complex vector space. Notation: $|\psi\rangle$
- Invented by Paul Dirac: "Bracket" → Bra + Ket

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

$$\langle\psi| = \begin{bmatrix} \alpha^* & \beta^* \end{bmatrix}$$

General Qubit State: $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$**:** $\alpha, \beta \in C$**.** $|\alpha|^2 + |\beta|^2 = 1$**.**

A Bra is the dual vector of a ket. **Notation:** $\langle\psi|$**.** It is the conjugate transpose of the ket.

Why do we need both ket and bra?

- Because Hilbert Space is inner product vector space: **Inner Products (between a vector and its dual)**
- **Inner product** $\langle\phi|\psi\rangle$ **gives:**
  - Overlap between two states
  - **A complex number**
  - **Probability amplitude of projecting one state onto another**

Kets, Bras, Operators

Example: Inner product $\langle \phi \mid \psi \rangle$ gives:

- $\langle 0 \mid \psi \rangle = \alpha \Longrightarrow$ Probability of measuring 0: $\mid \langle 0 \mid \psi \rangle \mid^2 = \mid \alpha \mid^2$
- **This is the measurement rule.**

The Hilbert Space:

- **Quantum states live in:** Complex vector spaces with inner products → **Hilbert space**
- **Properties:** Linear, Complex-valued, Norm defined via inner product with Normalization: $\langle \psi \mid \psi \rangle = 1$

Operators in Quantum Mechanics:

- **Operators are:** Linear transformations acting on kets
- **Matrix representation:** $A \mid \psi \rangle$
- **Example:** Pauli-X Gate: $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ acts as: $X \mid 0 \rangle = \mid 1 \rangle$, $X \mid 1 \rangle = \mid 0 \rangle$

$\Longrightarrow$ It is the **quantum NOT gate**.

Kets, Bras, Operators

Operators in Quantum Mechanics:

- **Operators are:** Linear transformations acting on kets, e.g.: $A \mid \psi\rangle$
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Types of Operators:

- **Unitary Operators: Used for quantum gates, with condition:** $U^\dagger U = I$**.**
  - **They preserve Norm and Probability** and ensure: $\langle\psi \mid \psi\rangle = 1$ after evolution
- **Hermitian Operators: Used for measurements, with Condition:** $A^\dagger = A$
  - **Properties:** Real eigenvalues and orthogonal eigenvectors (Proofs)
  - **Measurement outcomes = eigenvalues ($\Longrightarrow$ real numbers)**
  - **Expectation values:** $\langle A\rangle = \langle\psi \mid A \mid \psi\rangle$

Hermitian Operators: Proofs

- **Hermitian Operators: Used for measurables/measurements, with condition: $A^\dagger = A$**
  - **Eigenvalues of a Hermitian Operator Are Real**
    - Let's assume $|\psi\rangle \neq 0$ is an eigenvector: $A|\psi\rangle = \lambda|\psi\rangle$ where $\lambda \in C$ is a priori.
    - $\langle\psi|A|\psi\rangle = \lambda\langle\psi|\psi\rangle$
    - Using Hermiticity: $\langle\psi|A|\psi\rangle = \langle A^\dagger\psi|\psi\rangle = \langle A\psi|\psi\rangle$
    - Let's substitue: $A|\psi\rangle = \lambda|\psi\rangle \Longrightarrow$
    - $\langle\psi|A|\psi\rangle = \langle A^\dagger\psi|\psi\rangle = \langle A\psi|\psi\rangle = = \langle\lambda\psi|\psi\rangle = \lambda^*\langle\psi|\psi\rangle$
    - Equating: $\lambda\langle\psi|\psi\rangle = \lambda^*\langle\psi|\psi\rangle \Rightarrow \lambda = \lambda^*$
    - Therefore: $\lambda \in R$

Eigenvectors with Distinct Eigenvalues Are Orthogonal

- Let, $A|\psi\rangle = \lambda|\psi\rangle$ and $A|\phi\rangle = \mu|\phi\rangle$ with $\lambda \neq \mu$
- $\langle\psi|A|\phi\rangle = \mu\langle\psi|\phi\rangle$
- Using Hermiticity: $\langle\psi|A|\phi\rangle = \langle A\psi|\phi\rangle = \lambda\langle\psi|\phi\rangle$
- Therefore, $\lambda\langle\psi|\phi\rangle = \mu\langle\psi|\phi\rangle$
- $(\lambda - \mu)\langle\psi|\phi\rangle = 0$
- Since, $\lambda \neq \mu \Longrightarrow \langle\psi|\phi\rangle = 0$
- **Therefore, Eigenvectors corresponding to distinct eigenvalues are orthogonal.**

The Pauli Matrices

The Pauli matrices are three 2×2 Hermitian matrices:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \qquad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \qquad I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

The Pauli Set: $\{I, \sigma_x, \sigma_y, \sigma_z\}$

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

These matrices act on the computational basis states:

Pauli-X (Bit Flip)

$$\sigma_x|0\rangle = |1\rangle$$
$$\sigma_x|1\rangle = |0\rangle$$

Thus Pauli-X behaves like a quantum NOT gate.

Pauli-Z (Phase Flip)

$$\sigma_z|0\rangle = |0\rangle$$
$$\sigma_z|1\rangle = -|1\rangle$$

Pauli-Y (Bit + Phase Flip)

$$\sigma_y|0\rangle = i|1\rangle$$
$$\sigma_y|1\rangle = -i|0\rangle$$

- **Each Pauli matrix has eigenvalues:** $\lambda = \pm 1$
- **Pauli matrices are both Hermitian ($\sigma_i^\dagger = \sigma_i$) and Unitary ($\sigma_i^\dagger \sigma_i = I$). They correspond to observables and evolution (quantum gates).**
- **Each Pauli matrix squares to identity:** $\sigma_x^2 = \sigma_y^2 = \sigma_z^2 = I$ **i.e. applying the operation twice restores the original state e.g.:** $\sigma x(\sigma x \mid 0\rangle) = \mid 0\rangle$
- **Pauli matrices do not commute $\Longrightarrow$ Spin uncertainity relationship**
- **Basis for all single qubit operators i.e. any 2×2 Hermitian operator can be expanded as:** $A = a_0 I + a_x\sigma_x + a_y\sigma_y + a_z\sigma_z$ **i.e.**
  - **Any single-qubit Hamiltonian • Any density matrix • Any single-qubit observable can be written in the Pauli basis.**

Operators: Properties of the Unitary Matrices (Quantum Circuits)

1. **Linearity: Very fundamental**
    1. **A unitary operator U is linear:** $U(a \mid \psi\rangle + b \mid \phi\rangle) = aU \mid \psi\rangle + bU \mid \phi\rangle$ **for all complex numbers a , b.**
    2. **Why Linear? Because quantum states obey the superposition principle:** $\mid \psi_1\rangle \to U \mid \psi_1\rangle$ **and** $\mid \psi_2\rangle \to U \mid \psi_2\rangle$ **then their superposition must evolve as:** $a \mid \psi_1\rangle + b \mid \psi_2\rangle \to aU \mid \psi_1\rangle + bU \mid \psi_2\rangle$**.**
    3. **Linearity preserves superposition.**
2. **Invertibility:** Since, $U^\dagger U = I$. It implies: $U^{-1} = U^\dagger$. (Classical Gates might be irrevesible). Quantum Gates are reversible.
3. **Evolution is reversible. No information is destroyed $\Rightarrow$ Conservation of probability and Microscopic reversibility**
4. **Norm Preservation:** $\| U\psi \| = \| \psi \| \Longrightarrow \langle U\psi \mid U\psi\rangle = \langle \psi \mid \psi\rangle$. So total probability = 1 at all times.
5. **Inner Product Preservation:** Slightly Stronger than norm preservation: $\langle U\phi \mid U\psi\rangle = \langle \phi \mid \psi\rangle$.
6. This implies: Angles preserved Orthogonality preserved Transition probabilities preserved
7. **Continuity in Time:** $U(t+s) = U(t)U(s)$. $U(t) = e^{-iHt/\hbar}$ with H Hermitian.

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- Pauli matrices do not commute $\Longrightarrow$ Spin uncertainity relationship
- Basis for all single qubit operators i.e. any 2×2 Hermitian operator can be expanded as: $A = a_0 I + a_x \sigma_x + a_y \sigma_y + a_z \sigma_z$ i.e.
    - Any single-qubit Hamiltonian • Any density matrix • Any single-qubit observable can be written in the Pauli basis.
    - E.g.: The density matrix: $\rho = 1/2(I + r \cdot \sigma)$ where r is the Bloch vector
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- Rotations are generated by: $R_n(\theta) = e^{-i\theta\sigma_n/2}$
    - These are fundamental quantum gates. Also, $e^{i\theta(n\cdot\sigma)} = cos\theta . I + isin\theta . (n \cdot \sigma)$
- In QC, Pauli matrices correspond to the quantum gates
- Hamiltonians are represented in terms of them
- Quantum Error Correction: Errors are modeled as
    - X error (bit flip)
    - Z error (phase flip)
    - Y error (combined)

Kets, Bras, Operators

Projectors: Measurement Operator

- **Outer product:** $|\psi\rangle\langle\phi| \Longrightarrow$ This creates a **matrix (operator)**.
- For example: $|0\rangle\langle 0| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$. This is a projector.
- **Projectors: Measurement operator:** $P_0 = |0\rangle\langle 0|$
  - Probability of outcome: $\langle\psi|P_0|\psi\rangle$

Operators in Quantum Mechanics: Multi-Qubit States

- **Tensor product structure:** $|00\rangle = |0\rangle \otimes |0\rangle$ **;** $|01\rangle = |0\rangle \otimes |1\rangle$
- **Dimension doubles each time:**
  - 1 qubit → 2D
  - 2 qubits → 4D
  - n qubits → 2^n dimensional space

In fact, this exponential growth gives quantum advantage!

Key Takeaways

- A ket is just a complex vector.
- A bra is its conjugate transpose.
- Inner product gives probabilities.
- Operators are matrices.
- Quantum gates are unitary operators.
- Measurement operators are Hermitian.

Thank you